



Jacob Stuivenvolt-Allen

jsallen@ucar.edu | <https://jacobstuallen.github.io>

National Center for Atmospheric Research, Boulder, CO

Research Interests

Climate variability and change, ENSO dynamics, tropical-extratropical interactions, climate monitoring and prediction, satellite remote sensing, atmospheric circulation patterns, and large-scale climate phenomena.

Education

Ph.D. in Climate Science 2018–2022
Utah State University Logan, UT
Dissertation: Impact of the Changing Pacific on North American Drought, Atmospheric Rivers, and Explosive Cyclones
Advisor: Shih-Yu Simon Wang

B.S. in Environmental Science 2012–2016
Loyola Marymount University Los Angeles, CA

Professional Experience

Scientist IV 2025–Present
National Center for Atmospheric Research Boulder, CO

Postdoctoral Research Associate 2023–2025
Department of Earth and Planetary Sciences, Yale University New Haven, CT

Publications

Full publication and conference presentation list available on [Google Scholar](#)

In Preparation

Stuivenvolt-Allen, J., Eidhammer, T., McCrary R., Bukovsky, M., Rahimi, S., Chang, H., McGinnis, S. (2026). The North American CORDEX-CMIP6 WRF evaluation run: comparing historical simulations from 25 km to convection-resolving scales. *Journal of Advances in Modeling Earth Systems*.

In Review

Stuivenvolt-Allen, J., & Fedorov, A. (2025). Trade winds and climate sensitivity dictate the pace of warming in the eastern equatorial Pacific. *Science Advances*.

Published

- Stuivenvolt-Allen, J.**, & Fedorov, A. (2025). Atmospheric nonlinearity controls ENSO asymmetry in a hybrid statistical-dynamical model. *Geophysical Research Letters*. doi:10.1029/2025GL117456
- Stuivenvolt-Allen, J.**, Fedorov, A. V., Fu, M., & Heede, U. (2025). Widening of wind stress anomalies amplifies ENSO in a warming climate. *Journal of Climate*. doi:10.1175/JCLI-D-24-0126.1

11. Garuba, O. A., Liu, W., **Stuivenvolt-Allen, J.**, Balaguru, K., Hagos, S., & Wang, H. (2025). The Atlantic Meridional Overturning Circulation's Response to CO₂ Increase: Assessing the Roles of Surface Flux and Oceanic Advection Feedbacks. *Journal of Climate*. doi:10.1175/JCLI-D-24-0143.1
10. Johnson, Z., **Stuivenvolt-Allen, J.**, Mahan, H., Meyer, J., Miksch. (2023). Upper Colorado River streamflow dependencies on summertime synoptic circulations and hydroclimate variability. *Journal of Hydrometeorology*. doi:10.1175/JHM-D-23-0053.1
9. Dalanhese, L., **Stuivenvolt-Allen, J.**, LaPlante, M., Wang, S.-Y. (2023). An inter-comparison of South American extratropical cyclogenesis using Brazilian Navy data and ERA5. *International Journal of Climatology*. doi:10.1002/joc.8251
8. **Stuivenvolt-Allen, J.**, Wang, S.-Y., Chikamoto, Y., Meyer, J.D., Johnson, Z.F. and Deng, L. (2023). Growing Pacific linkage With Western North Atlantic Explosive Cyclogenesis. *Journal of Climate*. doi:10.1175/JCLI-D-22-0784.1
7. Wright, T., **Stuivenvolt-Allen, J.**, Affram, G., Hasan A., N., Ratterman, C., Zhang, W. (2023). Responses of extreme discharge to changes in surface air and dewpoint temperatures in Utah: Seasonality and Mechanisms. *Water*. doi:10.3390/w15040688
6. Pokharel, B., Sharma, S., **Stuivenvolt-Allen, J.**, Wang, S.-Y., LaPlante, M., Gillies, R. R., Khanal, S., Wehner, M. F., Rhoades, A. M., Hamal, K., Liu, W.-Y., Mukherjee, S., & Aryal, D. (2023). Amplified drought trends in Nepal increase the potential for Himalayan wildfires. *Climatic Change*. doi:10.1007/s10584-023-03495-3
5. **Stuivenvolt-Allen, J.**, Wang, S.-Y. (2022) North American fire weather catalyzed by the extratropical transition of tropical cyclones. *Climate Dynamics*. doi:10.1007/s00382-022-06561-1
4. **Stuivenvolt-Allen, J.**, Wang, S.-Y. S., Johnson, Z., & Chikamoto, Y. (2021). Atmospheric rivers impacting Northern California exhibit a quasi-decadal frequency. *Journal of Geophysical Research: Atmospheres*, 126, e2020JD034196. doi:10.1029/2020JD034196
3. Morgan, B., Spangler, K., **Stuivenvolt-Allen, J.**, Morrisett, C. N., Brunson, M. W., Wang, S.-Y. S., & Huntly, N. (2021). Water Availability for Cannabis in Northern California: Intersections of Climate, Policy, and Public Discourse. *Water*, 13(1). doi:10.3390/w13010005
2. **Stuivenvolt-Allen, J.**, Wang, S.-Y. S., LaPlante, M. D., & Yoon, J.-H. (2021). Three Western Pacific Typhoons Strengthened Fire Weather in the Recent Northwest U.S. Conflagration. *Geophysical Research Letters*, 48(3), e2020GL091430. doi:10.1029/2020GL091430
1. **Stuivenvolt-Allen, J.**, & Wang, S.-Y. S. (2019). Data-mining climate variability as an indicator of U.S. natural gas. *Frontiers in Big Data: Integrating Big Data with Earth System Models of Natural and Human Systems*. doi:10.3389/fdata.2019.00020

Selected Presentations

[Invited] Stuivenvolt-Allen, J., Eidhammer, T., McCrary R., Bukovsky, M., Rahimi, S., Chang, H., McGinnis, S. (2026). “How Far Can 12 km Take Us? Comparing historical simulations from 25 km to convection-resolving scales.” American Geophysical Union, New Orleans.

[Invited] Stuivenvolt-Allen, J., & Fedorov, A.V. (2025). “Atmospheric Nonlinearity Controls ENSO Asymmetry in a Hybrid Statistical-Dynamical Model” Princeton GFDL Seminar.

Stuivenvolt-Allen, J. (2025). “Impacts of the North American Circulation Response to Western North Pacific Tropical Cyclones” Seminar, Research Applications Laboratory at NSF NCAR.

Current Projects

North America CORDEX

- Running regional simulations of North American climate for historical and future emissions scenarios.
- Creating workflows for post-processing and sharing CF compliant regional model output.
- Comparing the benefits of updated simulations versus past modeling efforts.

Awards & Honors

- 2022:** Doctoral Researcher of the Year: Plants, Soils, and Climate Department, Utah State University
- 2021:** Top Award for Science Writing from the American Society of Journalists and Authors: The Butterfly and the Blaze
- 2020:** Graduate Enhancement Award, Utah State University
- 2019:** Best Research Presentation: Plant, Soils and Climate, Utah State University
- 2017:** Utah Master Naturalist: Wetlands Certification
- 2017:** Utah Master Naturalist: Mountains Certification
- 2016:** Sigma Xi National Research Honor Society Member

Professional Activities

Service

- Peer reviewer for GRL, ERL, JCLIM, CLIMD, JGR-A, etc.
- Mentor for NSF Significant Opportunities in Atmospheric Research and Science.
- Seminar organizer for Yale Department of Earth and Planetary Sciences (2022-2024).

Outreach and Communication

- Volunteer weather forecaster for Utah Public Radio (2018-2022)
- Scientific journalism:
 - Improving Our Relationship With Fire - An ARCGIS Storymap
 - The Butterfly and the Blaze - Eugene Weekly
 - Utah Water Issues - A Blog on Utah's Water Supply and Management
- Website development and climate monitoring: <https://jacobstuallen.github.io>

Technical Skills

Programming Languages: Python, R, MATLAB, IDL, NCL, Bash/Shell scripting, JavaScript

Global and Regional Models: CESM, E3SM, WRF, MPAS-A

Scientific Computing: Parallelized workflows, HPC environments, job schedulers (SLURM, PBS)

Development Tools: Git/GitHub, LaTeX, HTML/CSS, version control

References

Available upon request